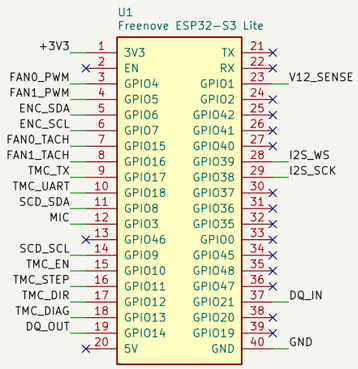
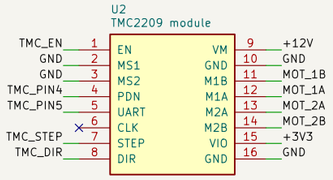
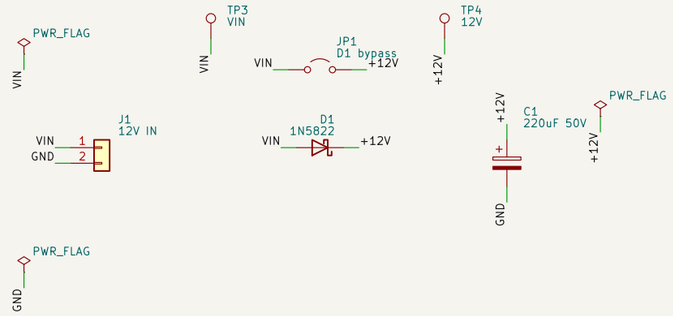
ESP32-S3 (Freenove Lite, FNK0099), soldered: USB toward the box's left wall



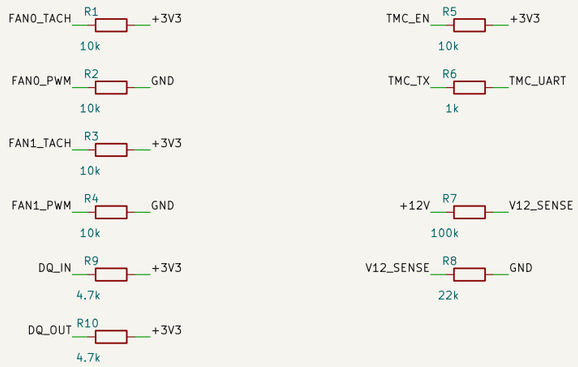
TMC2209 stepper driver (StepStick module), soldered



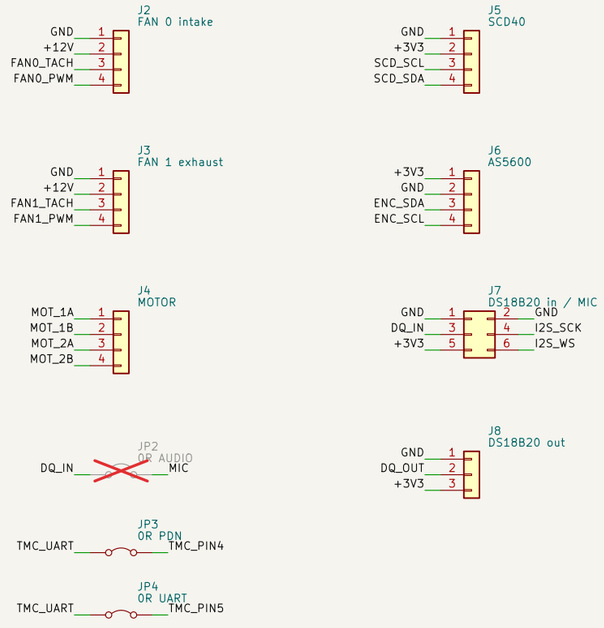
12 V input: D1 against reverse polarity; a jumper on JP1 bypasses it



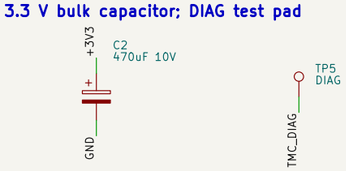
Fan, temperature-probe and driver resistors; 12 V sense divider



Connectors (2.54 mm male headers)



3.3 V bulk capacitor; DIAG test pad



Mounting holes (M2)



Nets with the same label are connected. Pins as in firmware/libs/WindowVent/src/Pins.h.

U1 pins 1–20 are its left header (3V3 … 5V) and 21–40 its right header (TX … GND), top to bottom as on the board.

U2: MS1 = MS2 = GND (UART address 0); CLK free. DIAG is not on this module's pins: GPIO 13 goes to TP5.

R6 1k: single-wire UART, TX through 1k and RX straight onto PDN_UART. JP3/JP4 (0R 0402, both fitted): the module's PDN_UART is on pin 4 or pin 5, depending on the maker.

R5 holds the driver off (EN high) while the ESP32 boots.

R7/R8: 12 V reads 2.16 V on GPIO 1 (ADC1). R2/R4 keep the fans slow at boot; R1/R3 tach pull-ups; R9/R10 1-Wire pull-ups.

J7 (2 x 3): row 1 (3V3, IN, GND) takes the indoor DS18B20 or the MAX9814; both rows take the INMP441 (I2S): SD on IN (GPIO 21), WS GPIO 39, SCK GPIO 38, L/R to the second GND.

AS5600: tie its DIR pin to GND at the module. C1 sits next to U2's VM pin.

JP2 (0R 0402 on the back, not fitted): fit it, or a wire, for the audio mode, a microphone on J7 read on GPIO 3 (GPIO 21 has no ADC); leave R9 out then.

Generated by tools/pcb_schematic.py from tools/pcb_parts.py
github.com/mmassico/window_air_vent

Sheet: /
File: window_vent_ctrl.kicad_sch

Title: Window vent controller

Size: A3	Date: 2026-11-07	**Rev: 1.3**
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